FYP EVALUATION DOCUMENT

CiteGuard

Automated Citation Integrity Checker for Academic Manuscripts

“Your references. Actually checked. Before you hit submit.”

Prepared for: Hamza
BS Software Engineering, 4th Year — Bahria University Islamabad

1. Problem Statement (Easy English)

When someone writes an academic paper, they reference (cite) many other papers to support what they're saying — sometimes hundreds of them. But almost nobody actually checks, one by one, if all those references are real and accurate before submitting. This causes real, common problems:

- Fake citations — referencing papers that don't actually exist, or that were later retracted
- Bad-journal citations — citing low-quality journals that don't do real peer review
- Wrong claims — saying a source proves something when it actually says something different or unrelated
- Padding — adding extra, irrelevant references just to look more "researched"

The result: mistakes and even fake references slip through into published research, because checking every single citation by hand takes too much time and effort for a person to do properly.

2. Proposed Solution (Easy English)

CiteGuard is a website where a researcher uploads their paper (PDF or Word file), and the system automatically:

- Finds every citation in the paper, both in the text and in the reference list
- Checks each one against real, trusted academic databases to confirm it actually exists
- Compares what the author claims a source says against what that source's abstract actually says, using AI, to catch mismatches
- Shows everything on a simple dashboard with a clear score, and lets the researcher download a full report

The whole point: catch every bad reference before submitting the paper, instead of a reviewer or editor catching it later, which is embarrassing and can hurt a researcher's reputation.

3. APIs Used — Free or Paid

API	What It's Used For	Cost
CrossRef	Confirms a citation is real, official DOI registry	Free — no API key required
OpenAlex	Confirms citations, covers 250M+ scholarly works	Free — no API key required
Semantic Scholar	Confirms citations, provides abstracts for the semantic check	Free — API access included
Unpaywall (suggested enhancement)	Checks if a source is legally free to read (open access)	Free
IEEE Xplore (optional, for engineering/CS papers)	Extra coverage for IEEE-published papers	Requires an institutional subscription (check if Bahria University has HEC Digital Library access)
Google Scholar	Not used — no official API exists at all	N/A — avoid this source entirely

4. Existing Products (Real Competitors, Checked)

Product	What It Does	Free or Paid
TrueCitation	Checks references against 17+ databases, flags fake/predatory sources	Free tier available
Recite / ReciteWorks	Checks reference-list accuracy for APA/Harvard style	Paid subscription
CiteMe	Citation verification, includes retraction checking via PubMed	Not fully confirmed

Product	What It Does	Free or Paid
Scholar Sidekick	Verification + retraction + open-access checking, with named sources	Not fully confirmed
SemanticCite (open-source)	Citation verification + AI semantic claim-checking — closest overlap to CiteGuard	Free, open-source (GitHub)
citecheck (open-source)	Verification + auto-repair suggestions for broken citations	Free, open-source

Honest note: this is a genuinely active space in 2026, not an empty gap. CiteGuard's real, defensible edge is being the only option that combines full local/private processing, semantic claim-checking, AND a dashboard built for non-technical researchers, all together — competitors have some of these, none have all three.

5. Tech Stack

Layer	Technology
Frontend / Dashboard	React
Backend	Python — FastAPI or Django
Document Parsing	PyMuPDF or GROBID
Semantic Claim-Checking	A local, lightweight NLP model (e.g. Sentence-BERT), no cloud AI API
Database	PostgreSQL
Report Generation	A PDF/HTML generation library (e.g. WeasyPrint)
Deployment	Self-hosted, Docker recommended

6. Recommended Enhancements

Enhancement	Why Add It	Effort
Retraction checking	A citation can be real but still retracted — changes the recommendation entirely; three competitors already do this	Low — Crossref already provides this data
Open-access classification	Tells the researcher if a source is legally free to read (Gold/Green/Hybrid/Bronze via Unpaywall)	Low — one more free API call
Auto-repair suggestions	Instead of just flagging a broken citation, suggest the likely correct fix	Medium-High — good Semester 2 stretch goal
API / MCP endpoint	Lets CiteGuard plug into other research tools/workflows, not just a standalone site	Medium — add once the core dashboard works

7. Hardware Check

A laptop with 16GB RAM is genuinely sufficient for developing and demoing CiteGuard. The document's "4GB VRAM" requirement refers specifically to graphics-card memory for running the local NLP model faster — a separate thing from general system RAM. If the laptop has no dedicated GPU, the NLP model simply runs on the CPU using regular RAM instead, which works fine for a smaller model like Sentence-BERT at FYP scale. This is not a blocker for the project.

8. Honest Feedback — Should This Be Your FYP?

Yes — this is a solid, legitimate, and technically substantial choice for a 4th-year Software Engineering FYP. Specific reasons:

- Real, documented problem: published research found citation error rates around 25% even in prestigious journals — this is not a made-up problem
- Genuine technical depth: document parsing, real API integration, and local NLP inference together are a legitimate systems-engineering challenge, not a single-trick demo
- Zero ongoing API costs: CrossRef, OpenAlex, and Semantic Scholar are all free with no key required, which removes a major cost/reliability risk other FYPs face
- Privacy-first design is a real, defensible engineering decision, not just a feature checkbox — it directly shapes your architecture (local NLP instead of a cloud AI API) and is a strong point to defend in a viva
- Feasible in two semesters: the core pipeline (parse → verify → semantic check → dashboard) is achievable, with retraction-checking and auto-repair as realistic stretch goals

The one real caution, worth repeating plainly: this is not an empty market. SemanticCite (open-source) does something very close to your semantic claim-checking feature already, and TrueCitation/Recite/CiteMe are established, live commercial products. This does not mean you shouldn't build it — it means your originality claim in the proposal needs to be specific (local-first deployment for Pakistani university adoption, a non-technical researcher dashboard, multi-style citation support, retraction + open-access enhancements) rather than a broad "nobody does this" claim, which a supervisor could disprove with the same research shown in this document.

Overall verdict: proceed with this as your FYP, with the honest positioning above built into your proposal from day one.